INCH-POUND

MIL-T-27/151C
8 June 2012
SUPERSEDING
MIL-T-27/151B
13 March 1991

MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN

Inactive for new design after 13 March 1991

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the product described herein shall consist of this document and the latest issue of specification MIL-PRF-27.

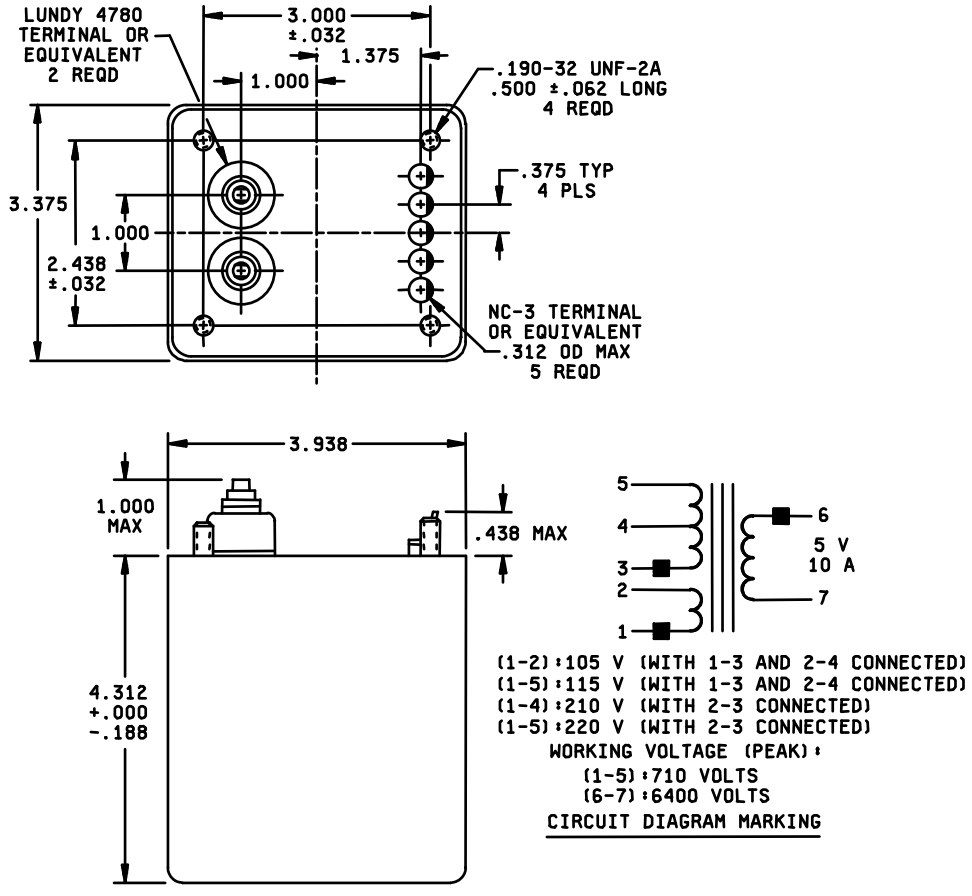

FIGURE 1. Dimensions and configuration.

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Inches	mm	Inches	mm
.032	0.81	.500	12.70
.062	1.57	1.000	28.40
.125	3.18	1.375	34.92
.188	4.78	1.438	36.53
.190	4.83	3.000	76.20
.312	7.92	3.375	85.72
.375	9.52	3.938	100.03
.438	11.13	4.312	109.52

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance on overall case dimensions is +.000, -.125 inch and tolerance on terminal positioning dimensions is ±.032 inch.
4. Marking shall be on the side of the case.

FIGURE 1. Dimensions and configurations – Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage:

(1-2): 105 volts, 50/60 Hz (with 1-3 and 2-4 connected).
(1-5): 115 volts, 50/60 Hz (with 1-3 and 2-4 connected).
(1-4): 210 volts, 50/60 Hz (with 2-3 connected).
(1-5): 220 volts, 50/60 Hz (with 2-3 connected).

Secondary voltage (6-7): 5 volts.

Secondary current (6-7): 10 amperes.

Voltampere: 50.

Working voltage (peak):

(1-5): 710 volts.
(6-7): 6,400 volts.

Design and construction:

Dimensions and configuration: See figure 1.

Duty cycle: Continuous.

Case: Metal encased.

Terminals: Solder lug:

(1, 2, 3, 4, and 5): NC-3 or equivalent.
(6 and 7): Lundy 4780 or equivalent.

Terminal height:

(1, 2, 3, 4, and 5): 0.438 inch, maximum.
(6 and 7): 1.000 inch, maximum.

Weight: 3.81 kg, maximum.

Operating temperature range: -55°C to +130°C.

Terminal strength: (1, 2, 3, 4, and 5): MIL-STD-202, method 211, test condition A, 5.0 pounds.
(6 and 7): MIL-STD-202, method 211, test condition E, 18 ounce-inches.

Dielectric withstanding voltage (at sea level):

(1-2): 2,000 V rms.
(3-5): 2,000 V rms.
(6-7): 10,000 V rms.

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Electrical characteristics:

No load: With 105 volts, 60 Hz across (1-2), current in (1-2) shall not exceed 350 mA and the power in (1-2) shall not exceed 6 watts.

Voltage across:
(3-5): 115 volts ±3 percent.
(6-7): 5.45 volts, ±3 percent.

Rated load: With 105 volts, 60 Hz across (1-2), the voltage across (3-5) and (6-7) shall be 115 volts ±3 percent and 5 volts ±3 percent, respectively.

DC resistance:
(1-2): 9.4 ohms, maximum.
(3-5): 11.1 ohms, maximum.
(6-7): 0.028 ohms, maximum.

Polarity: Additive, with terminals 2 and 3; and 5 and 6 connected.

Temperature rise: 40°C with 115 volts, rms, 60 Hz across (1-5) with (1-3) and (2-4) connected at an ambient temperature of 90°C.

VERIFICATION:

Qualification inspection: Not applicable.

Conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/151-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202

Custodians:
Army - CR
Navy - EC
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

(Project 5950-2009-091)

Review activities:
Army – AR, CR4
Navy – AS, MC, OS
Air Force - 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.daps.dla.mil

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